Project presentation on cybersecurity Awareness

Evaluating the Effectiveness of Cybersecurity Awareness Training in Organisations

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Abstract & Overview

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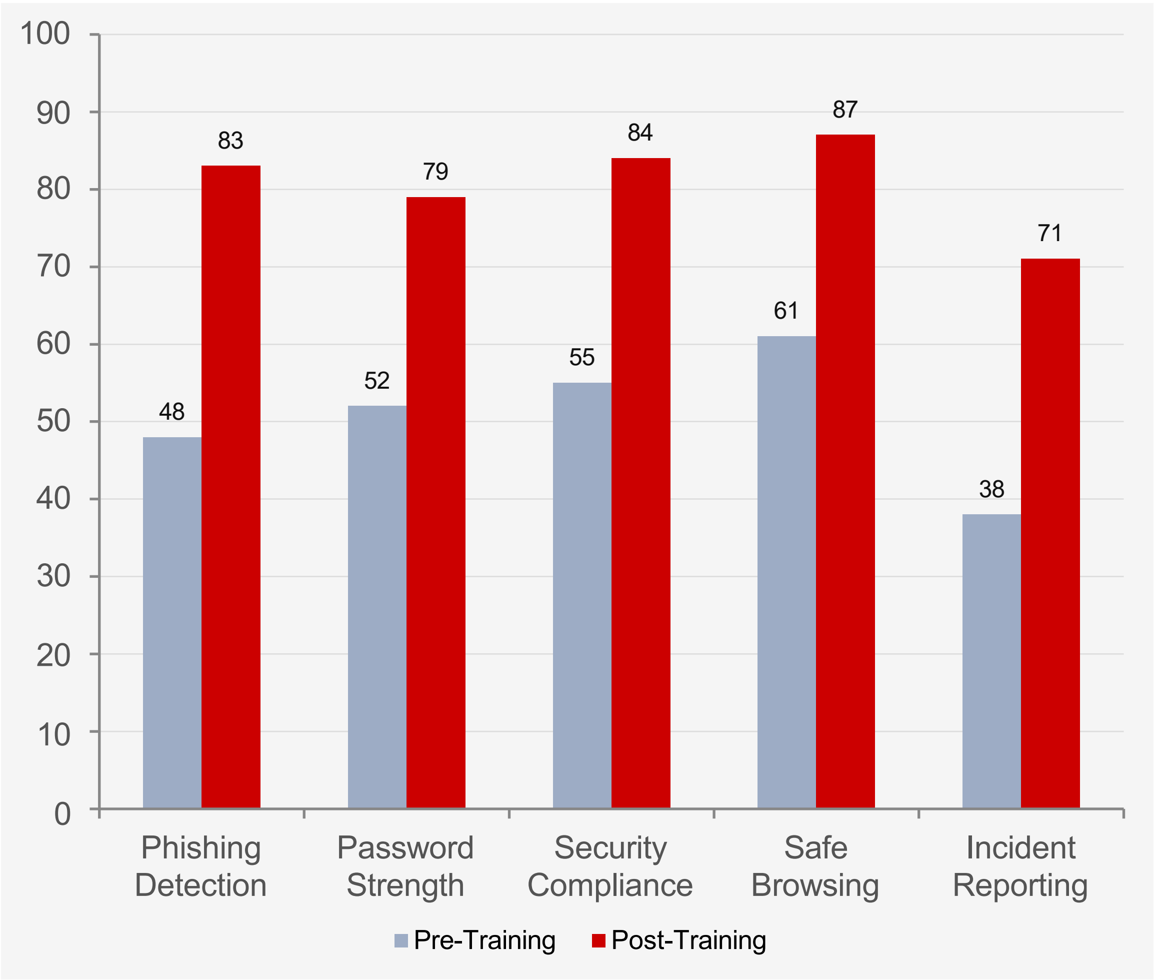
The Problem

- Human error accounts for 74% of all data breaches (Verizon DBIR 2023).
- Phishing is involved in >36% of all breaches — targeting people, not technology.
- Cybercrime will cost $10.5 trillion annually by 2025.
- CAT programmes widely adopted but little empirical evidence of effectiveness.
- Training format, frequency and delivery mode are under-studied.

This Study

- Mixed-methods longitudinal study across 6 months.
- 350-employee mid-sized technology services organisation.
- Pre & post assessments: phishing simulations, knowledge surveys, compliance audits, incident analysis.
- Compared 5 training modalities head-to-head.
- Statistical analysis via paired t-tests, ANOVA with Tukey HSD, effect sizes.

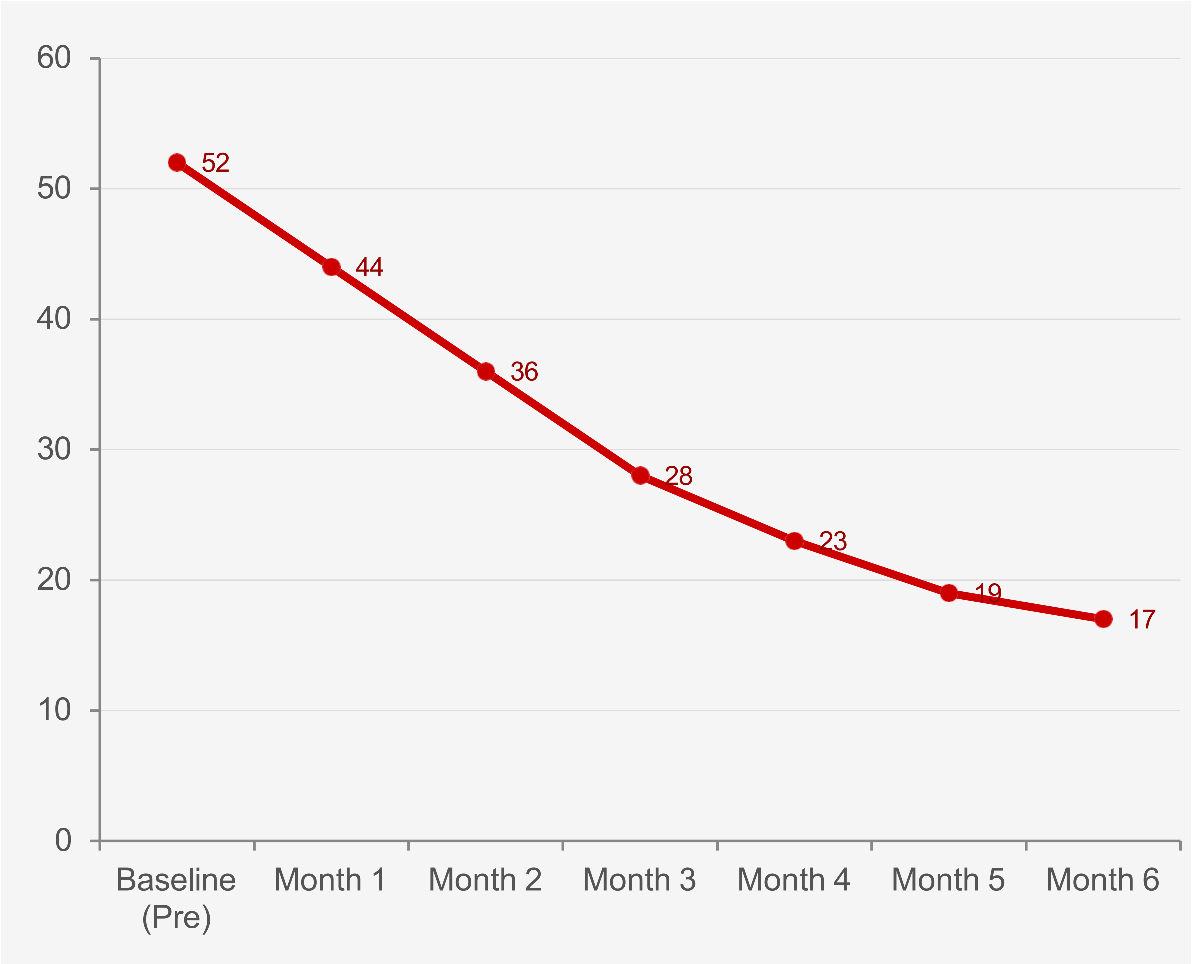
Pre- vs. Post-Training Performance



Statistical Summary

Metric	Δ pp	Cohen's d
Phishing Detection	+35	1.82
Password Strength	+27	1.45
Security Compliance	+29	1.61
Safe Browsing	+26	1.38
Incident Reporting	+33	1.74

Phishing Click Rate Reduction Over 6 Months



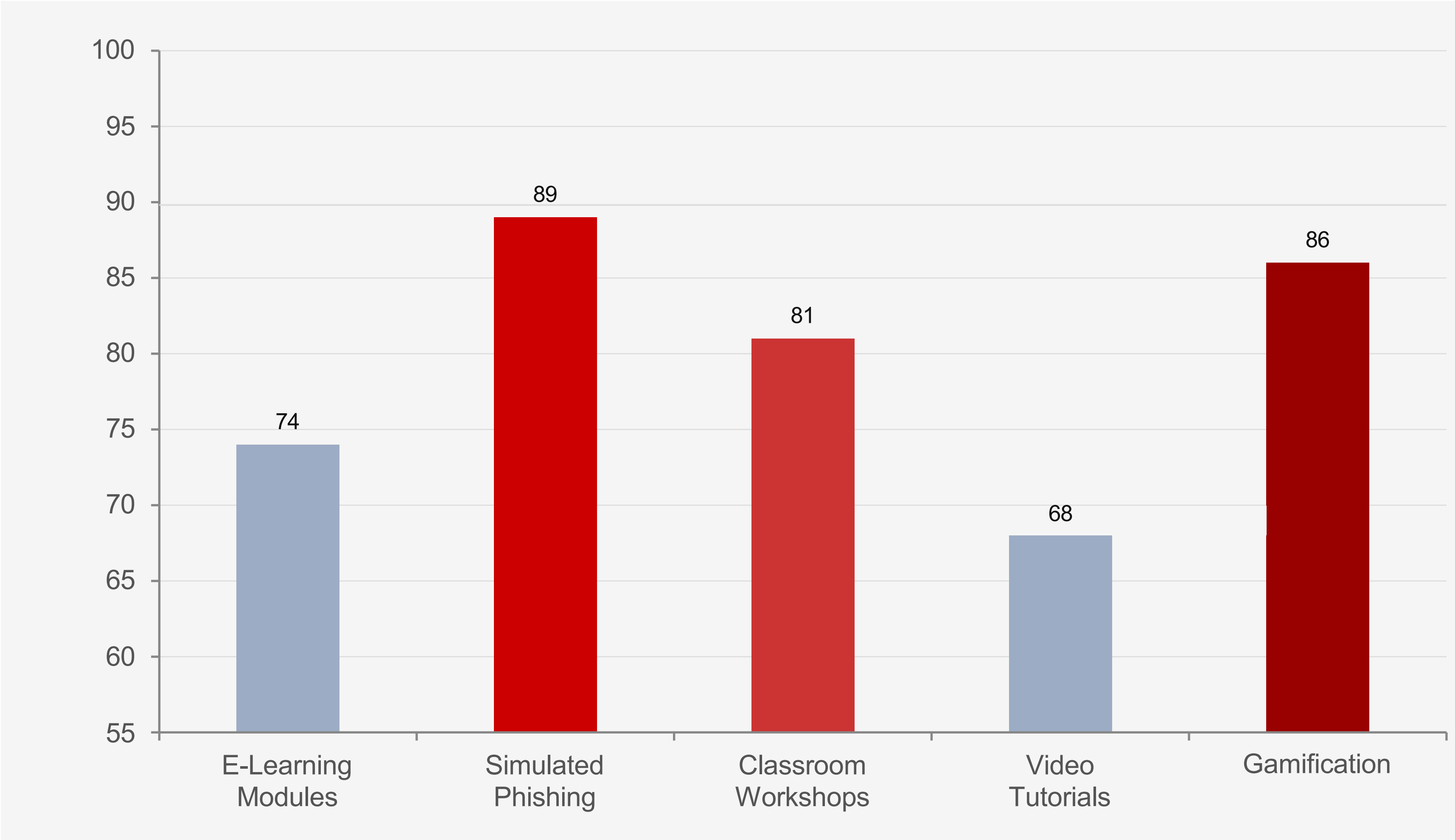
52%
Baseline click rate

17%
Month 6 click rate

67%
Relative reduction

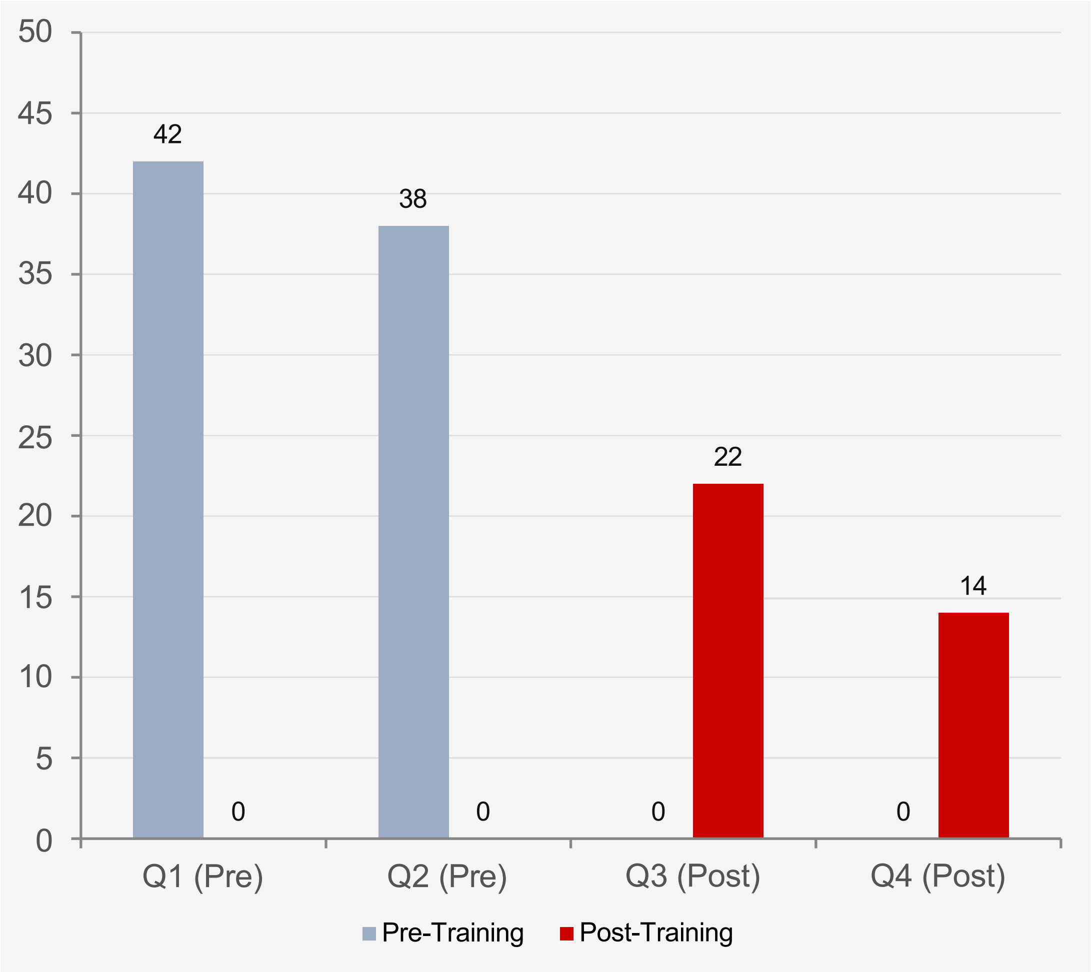
Months 1–2
Sharpest decline
(teachable moment effect)

Effectiveness of Training Delivery Methods



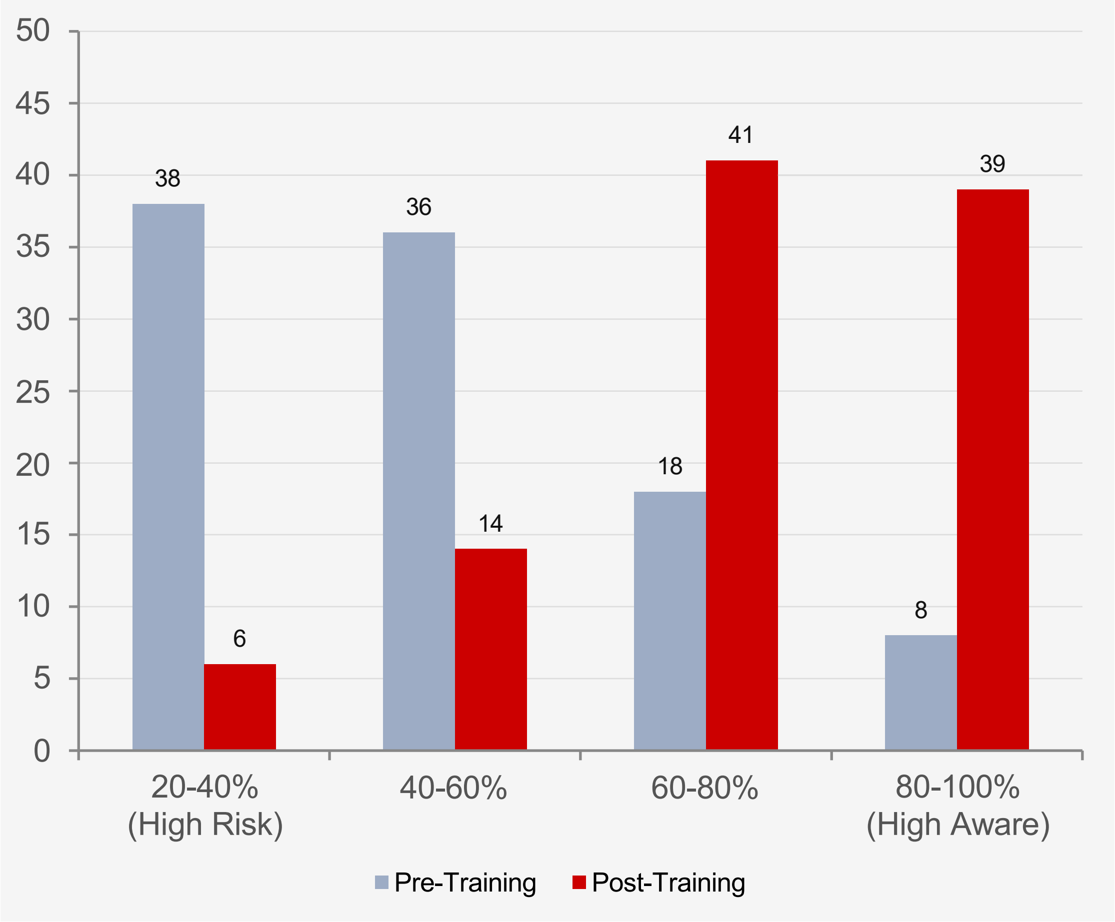
Reduction in Human-Attributable Security Incidents

Incident Type Breakdown



Incident Type	Pre	Post	↓%
Phishing Breaches	24.4	8.0	**67.2%**
Poor Passwords	10.8	4.5	**58.3%**
Unauth. Data Access	3.5	1.0	**71.4%**
Social Engineering	1.3	0.5	**61.5%**

Employee Awareness Score Distribution



Workforce Transformation

52.1%
Pre-Training Mean Awareness Score

79.3%
Post-Training Mean Awareness Score

38% → 6%
High-Risk Employees (scored <50%)

12.4% → 9.1%
Std Dev Reduction (lower variance

Discussion & Practical Implications

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Prioritise Active Training

Simulated phishing and gamification significantly outperform passive e-learning and video tutorials. Interactive, consequence-bearing training should be the core — not optional.

Shift to Continuous Training

Monthly micro-learning sustains behaviour change; annual events do not. Organisations should move to an always-on security awareness culture.

Target High-Risk Employees

Pre-training data revealed a long tail of highly vulnerable employees. Stratified training with intensive support for lowest performers delivers the best ROI.

Measure to Manage

Without pre/post measurement, organisations cannot calculate training value or identify vulnerabilities. Phishing simulations and compliance dashboards are integral, not add-ons.

Limitations & Future Research

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Limitations

Single Organisation

One mid-sized tech firm — generalisability to other industries/sizes is limited.

Hawthorne Effect

Employees aware of being measured may have performed better than training alone would produce.

Short Duration

6-month window cannot confirm long-term behaviour retention (12–24 month follow-up needed).

Simulated vs. Real

Simulated phishing may not match sophistication of targeted spear-phishing attacks, potentially overestimating resistance.

Future Research Directions

12–24 month longitudinal follow-up to assess long-term behaviour retention.

AI-powered adaptive training systems tailored to individual risk profiles and learning progressions.

Cross-organisational and cross-industry comparative studies for generalisable effectiveness metrics.

Conclusion

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+35 pp Phishing detection rate improvement (48% → 83%)

−67% Reduction in quarterly human-attributable security incidents

+29 pp Security policy compliance improvement (55% → 84%)

+33 pp Incident reporting rate improvement (38% → 71%)

#1 Simulated Phishing — most effective training delivery method (89%)

#2 Gamification — second most effective (86%), ahead of e-learning (74%)

Structured, multi-modal cybersecurity awareness training is an objectively effective and cost-efficient intervention for reducing human-based organisational risk.

Thank You

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